

SUMMARY

Computer Science student focused on distributed systems and live, event-driven software. Builder of *CodeCafé* (500+ GitHub stars) and open source contributor to *Apache Cassandra*. Skilled in full-stack development, backend engineering, and DevOps.

EDUCATION

- **Gettysburg College**

Bachelor of Science in Computer Science; Minor in Economics; Major GPA: 3.63/4.0

Honors: Dean’s List (2023, 2024); Presidential Scholar

Coursework: Computer Networks, Artificial Intelligence, OOP, Data Structures & Algorithms, Databases, Comp Org
- Gettysburg, PA
Expected Dec 2026

PROJECTS

Selected projects shown below; additional work is available on my portfolio site.

- **CodeCafé — Live Collaborative Cloud IDE (500+ ★, Hacker News front page)**

Creator & Sole Maintainer

Custom OT Engine (Core): Wrote a conflict-free **Operational Transformation (OT)** engine from scratch (like Google Docs); it handles real-time edits for 10+ concurrent users using Java Spring Boot, WebSockets, and Redis Lua scripts.

Cloud IDE Architecture: Built a browser-based workspace using React, TypeScript, and Monaco Editor that gives users an instant live preview of their HTML/CSS/JS code.

Containerized CI/CD: Dockerized the app and set up GitHub Actions to run tests (Jest/JUnit), build images, and automatically deploy the backend to AWS EC2 and frontend to Vercel.

Infrastructure: Set up Nginx on EC2 to handle secure WebSocket connections and TLS; used Cloudflare for DNS/CDN and Axios for secure data fetching.
- GitHub | Live | Demo
Jan 2025 – Present
- **Gedis — Redis Implementation & In-Memory Database Server**

Creator & Lead Developer

Concurrent TCP Server: Written in Go, this server handles multiple connections at once using goroutines and supports standard Redis commands like GET, SET, and DEL.

Thread-safe Storage: Implemented a key-value store with mutex locks and B-tree indexed sorted sets, allowing for fast range queries and safe concurrent access.
- GitHub
July 2025 – Present

OPEN SOURCE

- **Apache Cassandra**

Open Source Contributor

Database Enhancement: Added JSON output support for complex Cassandra settings, making it easier for external tools to read and configure the database.

Collaboration: Worked with maintainers to refine the design, fixed issues raised during code review, and added tests to verify the changes.
- View Merged Contribution
Aug 2025 – Present

EXPERIENCE

- **Software Engineer**

Poozle

Vector Recommendation Engine: Built a Rocchio-based recommendation filter on top of a Vector Database; aggregated negative user embeddings to exclude semantically similar job descriptions

Full Stack Development: Built a "Streaks" feature using lazy evaluation to track user progress; handled features across React Native and Node.js, including the Auto-Applier and general bug fixes.
- Remote
Oct 2025 - Mar 2026
- **Software Engineering Intern**

Gettysburg College

AI IDE Integration: Integrated a Gemini-powered assistant into DrJava to create a conversational coding tutor (similar to Cursor) that generates and edits code in real-time.

MCP Server & Tooling: Built a lightweight MCP server with Java HttpServer to analyze files and stream AI responses directly into the Java Swing front-end via SSE.

Academic Integrity Platform: Created a React app for proctored coding exams that tracks keystrokes, monitors window focus, and builds a timeline of student activity.
- Gettysburg, PA
May 2025 - Aug 2025

PROGRAMMING SKILLS

Languages	Java · Go · JavaScript · TypeScript · Python · Lua · SQL
Frameworks & Tools	React · Node.js · Spring Boot · FastAPI · React Native · JUnit · Docker
Databases & Storage	PostgreSQL · MySQL · Cassandra · Redis · Amazon S3 · ElastiCache
Concepts	Distributed Systems · Event-Driven Architectures · WebSockets · REST · gRPC